Decomposed Affordance Selection and Latent Generation for Physically Plausible Human-Scene Interaction

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Abstract

Language-conditioned human–scene interaction (HSI) requires motion to follow the instructed action while remaining compatible with complex 3D geometry. Existing methods commonly couple high-level interaction selection and frame-level motion synthesis within a single generator, which makes it difficult to preserve semantic intent throughout scene-conditioned execution. We present DeSeG, a two-stage framework that decomposes affordance selection from latent-conditioned motion generation. Its Planner fuses language, task goals, and a target-centered scene map into a compact affordance latent, while its diffusion Executor generates motion from this latent, a motion prefix, and body-centered local scene context. An occupancy-derived repulsive objective penalizes proximity to occupied regions during training without requiring gradient-based collision optimization at inference. Extensive experiments on LINGO show that DeSeG outperforms state-of-the-art HSI methods in motion quality and instruction alignment while markedly improving body–scene compatibility. Tests on a TRUMANS-derived subset and NC-Bench further demonstrate improved scene compatibility and instruction following.

1. Introduction

Language-conditioned human–scene interaction (HSI) aims to synthesize human motion that follows natural-language instructions in complex 3D environments [1, 15, 30]. Beyond producing realistic body dynamics, an HSI system must identify what interaction the instruction specifies, where it can be performed, and how the body should move with respect to surrounding geometry. This capability is central to embodied agents, virtual humans, and interactive content generation, where the same scene may support actions with distinct spatial and kinematic requirements.

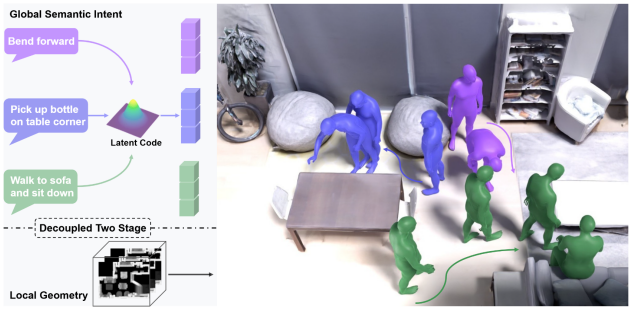


Figure 1. Representative human–scene interactions generated by **DeSeG**. The framework selects an affordance latent from the instruction, task goals, and target-centered scene geometry, and executes the interaction using body-centered local scene context.

A key challenge is to connect instruction-level intent with scene affordances. Existing scene-conditioned generators commonly fuse text, task goals, and scene features directly within a monolithic motion model. The resulting representation must simultaneously decide the intended interaction and resolve its frame-level realization. As tasks become longer and scenes support more diverse interactions, this motivates an explicit planning interface between interaction selection and dense motion synthesis. Such an interface should remain grounded in target geometry: the feasibility of sitting, lying, or kneeling depends not only on the action word, but also on the target location, object structure, and desired body configuration.

Scene-compatible execution presents a complementary challenge. A selected interaction must be transformed into continuous full-body motion while maintaining spatial coherence and avoiding severe body–scene penetration. Geometry-based optimization can refine generated motion at test time, but iterative guidance increases inference cost and complicates long-horizon generation. Incorporating geometric constraints during learning offers a more direct route to physically plausible execution while retaining effi-

cient diffusion sampling.

We propose **DeSeG**, a two-stage framework for decomposed affordance selection and latent-conditioned motion generation. The *Residual Affordance Planner* combines a language instruction, pelvis and hand goals, and a target-centered scene map to predict a compact affordance latent. The *Physics-Regularized Diffusion Executor* then synthesizes each motion segment from this latent, a short motion prefix, and a body-centered local scene crop. Separating these stages provides an explicit, geometry-aware interface between interaction planning and kinematic execution. During Executor training, an occupancy-derived repulsive objective penalizes joints that approach occupied regions without test-time gradient optimization.

Experiments on LINGO show that DeSeG achieves the best FID (2.882) and R-Precision (0.562) among methods evaluated under the original protocol while preserving motion diversity and attaining the lowest mean penetration (0.207). On a TRUMANS-derived subset, DeSeG further reduces FID by 31.4% and mean penetration by 48.1% relative to Lingo. We additionally extend NC-Bench with a paired affordance-conflict evaluation that complements conventional HSI metrics by measuring whether a model retains canonical interactions while following valid non-canonical instructions near the same object.

Our contributions are summarized as follows:

- We introduce a hierarchical HSI framework with a compact, language- and geometry-grounded affordance latent that conditions scene-aware motion generation.
- We develop an occupancy-derived repulsive objective that penalizes proximity to occupied regions during diffusion training, avoiding collision-gradient updates at inference.
- We demonstrate improved semantic alignment and geometric quality on LINGO and a TRUMANS-derived subset, and introduce a paired affordance-conflict benchmark for analyzing interaction selection.

2. Related Work

2.1. Human–Scene Interaction Synthesis

Text-conditioned motion generation has progressed rapidly through latent-variable and diffusion models [1, 2, 5, 8, 10, 29]. These methods synthesize diverse, semantically aligned motion, but generally assume free space. Human–object interaction methods introduce object geometry and contact constraints [25], although they often focus on short, atomic interactions with individual objects.

Human–scene interaction extends motion synthesis to large, cluttered 3D environments. HUMANISE [30] establishes a language-grounded HSI setting by aligning motion, text, and scanned scenes, while Move-as-You-Say [31] further grounds motion in scene affordances. PROX [12] optimizes static human poses in 3D scenes, whereas SceneDiffuser [14], Lingo [15], and TRUMANS [16] synthesize scene-conditioned motion and long-horizon behaviors. Recent systems increasingly adopt structured generation. SceMoS [7] combines a geometry-grounded global planner with local motion generation, and HOSIG [35] composes grasping, navigation, and trajectory-conditioned synthesis for dynamic-object interaction. DeSeG complements these directions with a target-centered affordance Planner and a body-centered diffusion Executor, using a compact latent to connect interaction selection with local scene-conditioned motion. In contrast to planners that predict only global routes, the DeSeG Planner encodes the 3D geometry surrounding an interaction target together with language and body goals. The Planner represents locomotion endpoints and object-centered actions; the Executor updates its body-centered occupancy crop at every rollout.

2.2. Latent Representations for Human Motion

Latent representations provide compact interfaces for motion generation. VAE-, VQ-VAE-, and Transformer-based approaches encode motion into continuous or discrete spaces for reconstruction, generation, and editing [9, 24, 38]. MotionCLIP [28] aligns motion and CLIP representations to support language control and zero-shot transfer. In scene-conditioned synthesis, however, the latent must also reflect spatial goals and the geometry that supports an interaction.

Hierarchical representations make this interface explicit. SemGeoMo [4] predicts contact geometry before synthesizing human–object motion. DeSeG instead learns a residual conditional latent from language, pelvis and hand goals, and a target-centered scene map. This latent serves as an affordance-level condition for a separate scene-aware diffusion model. The Planner learns interaction selection before the Executor is trained to generate motion from the frozen planning representation.

2.3. Physics-Aware Human Motion Synthesis

Physics-aware motion generation follows two main paradigms. Simulator- and reinforcement-learning-based methods enforce dynamic feasibility through physical interaction [23, 32, 37], but their control and optimization pipelines can be difficult to integrate with large generative models. Geometry-guided methods instead use signed-distance or contact objectives during sampling and refinement [17, 39, 40]. Although effective, repeated test-time optimization increases computational cost.

Training-time geometric objectives offer an alternative. SceneDiffuser [14] incorporates scene constraints into conditional diffusion, while potential-field formulations provide differentiable penalties near occupied regions [34]. DeSeG applies an occupancy-derived repulsive potential to predicted clean joints during diffusion training. This

objective shapes the learned generator without requiring collision-gradient updates during inference.

3. Method

As illustrated in Fig. 2, **DeSeG** consists of a **Residual Affordance Planner** and a **Physics-Regularized Diffusion Executor**. The Planner represents the interaction selected from language, task goals, and target-centered geometry as a compact latent. The Executor converts this latent into continuous motion using a short motion prefix and body-centered scene context.

3.1. Problem Formulation

Given a language instruction $\mathcal{T} = \{w_1, \ldots, w_L\}$, a 3D scene $\mathcal{S}$, task goals $\mathbf{g}$, and an initial motion prefix $\mathbf{M}_{pre}$, HSI synthesis learns the conditional distribution

$$p(\mathbf{M} \mid \mathcal{T}, \mathcal{S}, \mathbf{g}, \mathbf{M}_{pre}), \tag{1}$$

where the output $\mathbf{M} \in \mathbb{R}^{T\times J\times 3}$ contains J=28 3D body joints over T frames in a y-up coordinate system. Each instruction describes one action segment, and long-horizon motion is composed autoregressively. SMPL-X [19, 21] parameters are recovered from the predicted joints using a learned joints-to-body regressor. The task goals comprise a pelvis target and optional left- and right-hand targets, together with masks specifying which goals are active. They determine the desired body placement for both interaction and locomotion segments.

We introduce an affordance latent $\mathbf{z}$ between interaction planning and dense motion generation. Let $\mathcal{V}_g = \Psi_g(\mathcal{S}, \mathbf{g})$ denote the target-centered scene map used by the Planner and $\mathcal{V}_k$ the body-centered scene crop at rollout k. The Planner computes a deterministic condition, and the diffusion Executor samples a continuation:

$$\begin{aligned} \mathbf{z}_k &= f_\phi(\mathcal{T}, \mathbf{g}, \mathcal{V}_g) = \mu_\phi, \\ \mathbf{M}^{(k)}_{new} &\sim p_\varphi(\cdot \mid \mathbf{M}^{(k)}_{pre}, \mathbf{z}_k, \mathcal{V}_k). \end{aligned} \tag{2}$$

Here f_ϕ is the Planner's prior-mean prediction, and p_φ is the diffusion model. The conditional Gaussian introduced below supervises Planner training; it is not sampled by the Executor. Both stages use heading-aligned scene maps: $\mathcal{V}_g$ describes geometry near the task target, whereas $\mathcal{V}_k$ follows the body. The generated continuation supplies the prefix for the next rollout.

3.2. Residual Affordance Planner

Inspired by hierarchical vision–language–action methods [33], the Planner uses a residual conditional variational autoencoder (CVAE) [18, 27] to learn a conditional affordance prior. A future-motion encoder and an auxiliary decoder provide motion supervision during Planner training.

Future-Motion Encoder. During training, an MLP encodes the ground-truth future states $\mathbf{S}_{t+1:t+M}$ and predicts a residual posterior mean $\Delta\mu_\theta$ and standard deviation σ_θ. This branch guides the latent toward the demonstrated future motion and is removed after Planner training.

Target-Centered Geometric Perception. The scene is represented as a 3D occupancy grid. For each task, we extract a map around the target and express it in a heading-aligned coordinate frame. A frozen CLIP encoder [26] produces a pooled language feature, which is concatenated with the pelvis and hand goals to form the Planner query. A 3D-CNN [20] converts the target-centered occupancy map into spatial tokens. Cross-attention retrieves the geometry relevant to the requested interaction, and MLP heads predict the prior parameters μ_ϕ and σ_ϕ. The resulting latent jointly summarizes the instructed action, body goals, and geometry surrounding the target.

Let $\mathbf{f}_T$ be the pooled CLIP feature and $\mathbf{f}_g$ the embedded task goals. The fused query $\mathbf{q} = W_q[\mathbf{f}_T; \mathbf{f}_g]$ attends to the map tokens $\mathbf{H}_g = E_{3D}(\mathcal{V}_g)$:

$$\mathbf{h}_{aff} = \mathrm{softmax}\left(\frac{\mathbf{q}(W_k\mathbf{H}_g)^\top}{\sqrt{d}}\right) W_v\mathbf{H}_g. \tag{3}$$

The attended geometry feature is concatenated with the language–goal query to predict the prior parameters.

Residual Latent Space. During Planner training, the conditional prior is

$$p_\phi(\mathbf{z} \mid \mathcal{T}, \mathbf{g}, \mathcal{V}_g) = \mathcal{N}(\mu_\phi, \sigma^2_\phi). \tag{4}$$

The future-motion encoder predicts a residual shift rather than an independent posterior mean, yielding $q_\theta(\mathbf{z}) = \mathcal{N}(\mu_\phi + \Delta\mu_\theta, \sigma^2_\theta)$. Reconstruction encourages the sampled posterior code to represent the demonstrated continuation, the KL term aligns this posterior with the conditional prior, and latent alignment matches μ_ϕ to the stopped-gradient posterior mean. A training-only auxiliary decoder reconstructs future motion from the sampled posterior code and the current state.

The Planner objective combines these three terms:

$$\mathcal{L}_{plan} = \lambda_{rec}\mathcal{L}_{rec} + \beta\, D_{KL}(q_\theta \| p_\phi) + \lambda_{lat}\mathcal{L}_{lat}. \tag{5}$$

$\mathcal{L}_{rec}$ supervises future-motion reconstruction, whereas $\mathcal{L}_{lat} = \|\mu_\phi - \mathrm{sg}(\mu_\phi + \Delta\mu_\theta)\|^2_2$ aligns the prior and posterior means, where sg stops the gradient. After Planner training, the future-motion encoder and auxiliary decoder are discarded.

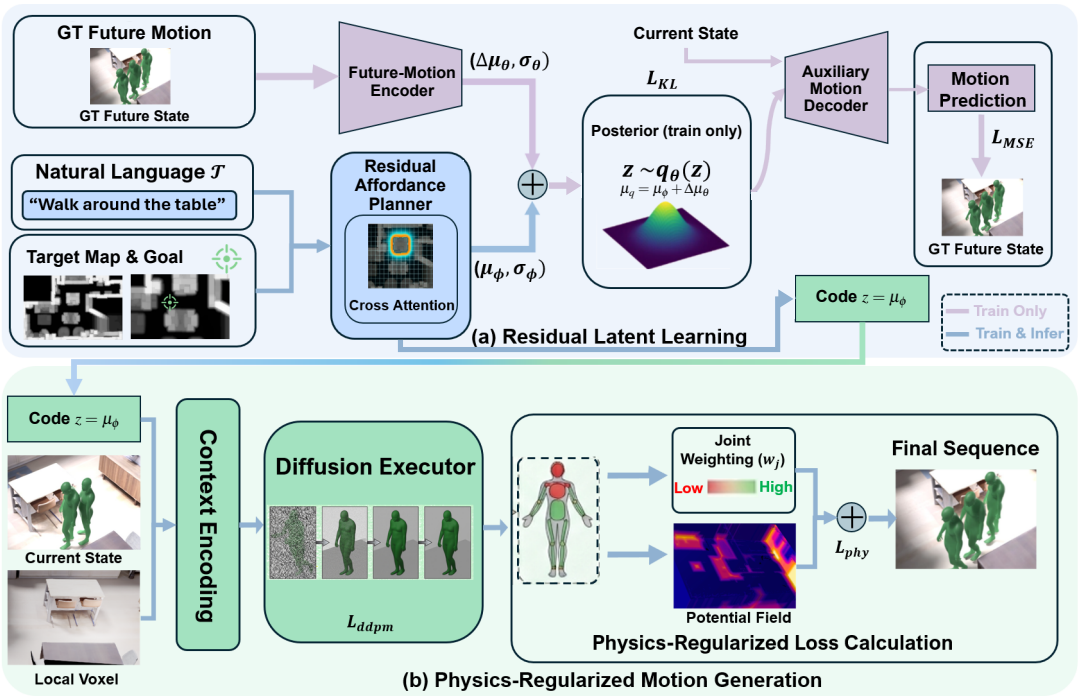


Figure 2. **Overview of DeSeG.** The Residual Affordance Planner fuses language, task goals, and a target-centered scene map. The Physics-Regularized Diffusion Executor uses the Planner's deterministic prior mean with a motion prefix and body-centered scene context to generate the next segment. The future-motion encoder, posterior sampling, and auxiliary decoder are used only during Planner training. The lower panel illustrates the training-time repulsion objective, not an inference-time refinement step.

3.3. Physics-Regularized Diffusion Executor

The Executor translates the affordance code $\mathbf{z}$ into a dense motion sequence $\mathbf{M}$ conditioned on local scene geometry. We use a Transformer-based diffusion model [13, 22] and add a training-time repulsion objective to discourage penetration.

Conditional Diffusion Backbone. Given a clean segment $\mathbf{M}_0$, the forward process samples

$$\mathbf{M}_t = \sqrt{\bar{\alpha}_t}\mathbf{M}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, I), \tag{6}$$

and trains the denoiser with

$$\mathcal{L}_{ddpm} = \mathbb{E}_{t,\boldsymbol{\epsilon}}\left[\ell_{\text{smooth-}L_1}\left(\boldsymbol{\epsilon}, \epsilon_\varphi(\mathbf{M}_t, t, C)\right)\right]. \tag{7}$$

The loss is evaluated on coordinates outside the fixed motion prefix. The corresponding reverse process is $p_\varphi(\mathbf{M}_{t-1} \mid \mathbf{M}_t, C)$. The context C comprises the deterministic Planner mean $\mathbf{z} = \mu_\phi$ and a body-centered scene voxel $\mathcal{V}_k$. Scene and latent tokens condition a Transformer denoiser through self-attention. Each autoregressive rollout preserves a short motion prefix and generates its continuation; the local frame and scene crop are then updated around the body for the next segment. For long-range locomotion, an optional external controller can use A* [11] to provide coarse waypoints, followed by local collision-aware steering and action scheduling. These controllers are external to the learned model. NC-Bench disables route planning and scheduling for every method. Interaction segments use the same backbone and follow their registered pelvis and hand goals directly. The same Executor handles locomotion and object-centered interaction, conditioned on the Planner's affordance latent and the body-centered scene crop.

Autoregressive Scene-Conditioned Rollout. At the beginning of each rollout, the motion prefix defines a body-centered coordinate frame. We transform the task goals and nearby occupancy crop into this frame, denoise a motion segment, and map the generated continuation back to world coordinates. The segment boundary then initializes the next rollout. For an interaction segment, the Planner map is centered at its task goal, whereas the Executor crop follows the moving body. The maps and goals are expressed in the rollout's heading-aligned frame. Multi-action sequences use an external scheduler to supply successive action instructions and task goals.

Occupancy-Derived Repulsive Loss. Inspired by the Humanoid Potential Field [34], we add an occupancy-derived repulsion term to the diffusion training objective.

Since the network predicts noise ϵ_φ, Tweedie's formula [3, 6] estimates the clean motion $\hat{\mathbf{M}}_0$ at diffusion step t:

$$\hat{\mathbf{M}}_0 = \frac{\mathbf{M}_t - \sqrt{1-\bar{\alpha}_t}\cdot\epsilon_\varphi(\mathbf{M}_t, t, C)}{\sqrt{\bar{\alpha}_t}}. \tag{8}$$

We extract joint coordinates $o_{j,k} \in \mathbb{R}^3$ from $\hat{\mathbf{M}}_0$ and compute a distance representation $\mathcal{D}$ from the corresponding local occupancy crop. The repulsive potential is

$$U(d) = \begin{cases} \xi(1-2d/d_0), & d < 0, \\ \xi(1-d/d_0)^2, & 0 \le d \le d_0, \\ 0, & d > d_0. \end{cases} \tag{9}$$

Here d_0 controls the range and ξ scales the penalty. Grid construction, interpolation, and loss weights are specified in the supplementary material.

The weighted penalty is

$$\mathcal{L}_{phy}(\hat{\mathbf{M}}_0) = \frac{1}{T}\sum_{k=1}^{T}\sum_{j=1}^{J}\bar{w}_j \cdot \widetilde{U}(o_{j,k}), \tag{10}$$

Here $\bar{w}_j = w_j / \sum_i w_i$ and $\widetilde{U}$ denotes the interpolated grid penalty. The auxiliary term is applied only at diffusion indices $t \le 5$. Interpolation makes the loss differentiable with respect to joint locations.

Optimization Strategy. We first train the Planner with its reconstruction, KL, and latent-regression objectives. We then freeze its weights and train the Executor using $\mathbf{z} = \mu_\phi$. The same deterministic interface is used at inference.

The Executor objective is

$$\mathcal{L}_{overall} = \mathcal{L}_{ddpm} + \lambda\,\mathbb{E}_{t,\boldsymbol{\epsilon}}\big[\mathbf{1}_{t\le 5}\mathcal{L}_{phy}(\hat{\mathbf{M}}_0)\big], \tag{11}$$

where λ weights the geometric penalty. Samples with $t > 5$ contribute zero repulsion to the batch average. Posterior motion supervision is confined to the first training stage; Executor training and inference both use the same prior-mean condition. At inference, diffusion noise supplies motion variation, and neither the potential field nor its gradients are evaluated.

4. Experiments

We evaluate DeSeG on motion quality, text alignment, scene compatibility, and affordance selection. The standard LINGO evaluation in Table 1 provides the primary comparison with existing HSI methods, while NC-Bench in Table 2 analyzes the role of hierarchical planning under competing language and object cues.

4.1. Experimental Setup

Dataset and Preprocessing. We train DeSeG on LINGO [15], which contains language-annotated human motion in large indoor scenes, and follow its scene-level train/test split. Scenes are represented by 3D occupancy grids. Target-centered maps are extracted for affordance planning, whereas body-centered crops are updated during motion execution. The same occupancy representation is converted online into the distance field used by the repulsive training objective.

Evaluation Metrics. Table 1 reports conventional HSI metrics: Fréchet Inception Distance (FID), Diversity, Multi-modality, R-Precision [29], and mean/maximum scene penetration. FID measures the distance between generated and real motion distributions; Diversity and Multi-modality measure global and instruction-conditioned variation; R-Precision evaluates text–motion retrieval; and penetration measures geometric overlap between the generated body and the scene. We additionally report PFC and contact ratio (Cont.) on the TRUMANS-derived subset.

Baseline Setup. We compare with Lingo [15], TRUMANS [16], TeSMo [36], and SceMoS [7]. All methods in the upper block of Table 1 are trained or retrained on LINGO using the same scene split and are evaluated with the original LINGO protocol. In particular, SceMoS is retrained on LINGO rather than evaluated through cross-dataset transfer. HOSIG [35] focuses on dynamic-object manipulation and is outside the static-scene setting considered here. Component variants use the available trained configurations of DeSeG.

Implementation Details. The Planner combines frozen CLIP features, a 3D scene encoder, and cross-attention to produce an affordance latent. A Transformer diffusion model executes motion from the deterministic Planner mean and body-centered scene context, with diversity arising from diffusion noise. The two stages are trained sequentially, and the Planner is frozen during Executor training. Architecture and optimization details are provided in the supplementary material.

Negative Constraint Benchmark. NC-Bench contains 30 Lingo-based cases: six preregistered object anchors for each of bed, sofa, chair, toilet, and bathtub. Every case has a canonical task and an affordance-conflict task. A pair shares the scene, object anchor, two-frame prefix, world transform, initial heading, conditioning masks, duration, camera, and diffusion seed. The canonical task uses the recorded interaction goal; the conflict task uses a nearby occupancy-

Table 1. **Quantitative comparison on LINGO and a TRUMANS-derived subset.** Bold denotes the best result under the original LINGO protocol. All methods in the upper block are trained on LINGO and evaluated under the same protocol. The lower block reports zero-shot transfer of LINGO-trained models to the TRUMANS-derived subset.

Method	**FID**↓	**Diversity**→	**M-modality**→	**R-Precision**↑	**Pene$_{\text{mean}}$**↓	**Pene$_{\text{max}}$**↓
Comparison with baselines						
TRUMANS [16]	3.286	4.984	3.036	0.413	0.404	1.825
TeSMo [36]	3.558	5.328	2.956	0.392	0.278	**0.915**
Lingo [15]	3.093	5.832	**2.239**	0.437	0.392	1.229
SceMoS [7]	3.147	5.095	2.403	0.512	0.359	1.194
Available trained configurations (not isolated component ablations)						
w/o Hierarchical Latent	3.279	5.461	2.439	0.430	0.241	0.938
w/o Cross-Attention	3.113	5.636	2.721	0.457	0.239	0.946
w/o Physical Regularization	2.983	5.822	2.524	0.514	0.367	1.393
DeSeG (full)	**2.882**	**5.865**	2.541	**0.562**	**0.207**	0.921
Comparison on TRUMANS	**FID**↓	**Diversity**→	**PFC**↓	**Cont**↑	**Pene$_{\text{mean}}$** ↓	**Pene$_{\text{max}}$** ↓
Lingo [15]	0.493	2.932	0.657	0.941	2.279	13.376
DeSeG	0.338	2.853	0.621	0.986	1.183	9.187

validated goal for the instructed action. Route planning and action scheduling are disabled during this evaluation.

The five object categories support complementary canonical and conflict actions: lying versus kneeling for beds, sitting versus standing for sofas, sitting versus squatting for chairs, sitting versus standing for toilets, and showering versus kneeling for bathtubs. All instructions occur in the LINGO training vocabulary. Object anchors are selected before generation, and each canonical/conflict pair is rendered with three shared random seeds. This design tests whether the same scene-conditioned generator can preserve an object's standard interaction and execute a different valid instruction nearby.

We recover each object anchor and canonical goal from the endpoint of a real LINGO episode. The conflict goal is selected in free space 0.4–1.0 m from the anchor toward the initial body position and is validated against scene occupancy. Both conditions begin with the same heading toward the object. Their durations are matched to the source interaction and quantized to the model's rollout interval. Conflict prompts are encoded through the same online CLIP pathway as canonical prompts rather than through a dataset index.

We compare DeSeG with a matched-depth **Monolithic+Repulsion** model using the same geometric objective, a legacy DeSeG variant without physical regularization, and Lingo. Semantic–Geometric Consistency (SGC) reports blinded human judgments of instructed-action agreement in conflict clips; spatial task completion is assessed separately by automatic success. The latter requires the generated action to match the instruction, a terminal pelvis error at most 0.3 m, and a valid complete motion. **Can. Succ.** and **Conf. Succ.** report success on the two conditions; **Paired Succ.** requires both motions in a matched pair to succeed; and **Prior Capture** measures how often a conflict clip reverts to the object's canonical action. Prompts and the paired evaluation protocol are detailed in the supplementary material.

Monolithic+Repulsion matches the Executor architecture, motion representation, sampling protocol, and occupancy-derived repulsion, but conditions a single diffusion network directly instead of using the affordance latent. For condition $x \in \{can, conf\}$, we compute

$$S_x = A_x \, G_x \, V_x \, R_x, \tag{12}$$

where all four factors are binary indicators. A_x indicates a match to the registered action, G_x a terminal pelvis error of at most 0.3 m, V_x a valid complete motion, and R_x support by the action evaluator. An unsupported or invalid output fails success and is not counted as Prior Capture. Paired success is $S_{can}S_{conf}$ for the same case and seed: both motions must satisfy the action, goal, and validity criteria.

The automatic action evaluator recognizes six classes—kneel, lie, shower, sit, squat, and stand—and is trained on real LINGO episodes disjoint from the benchmark sources. For SGC, 30 participants each judge 20 conflict clips, yielding 600 responses. Every case–method cell receives five judgments, methods are balanced within participants, and no participant views the same case twice. SGC is the proportion of action judgments matching the conflict instruction. Automatic metrics average the three seeds within each case; Table 2 reports point estimates.

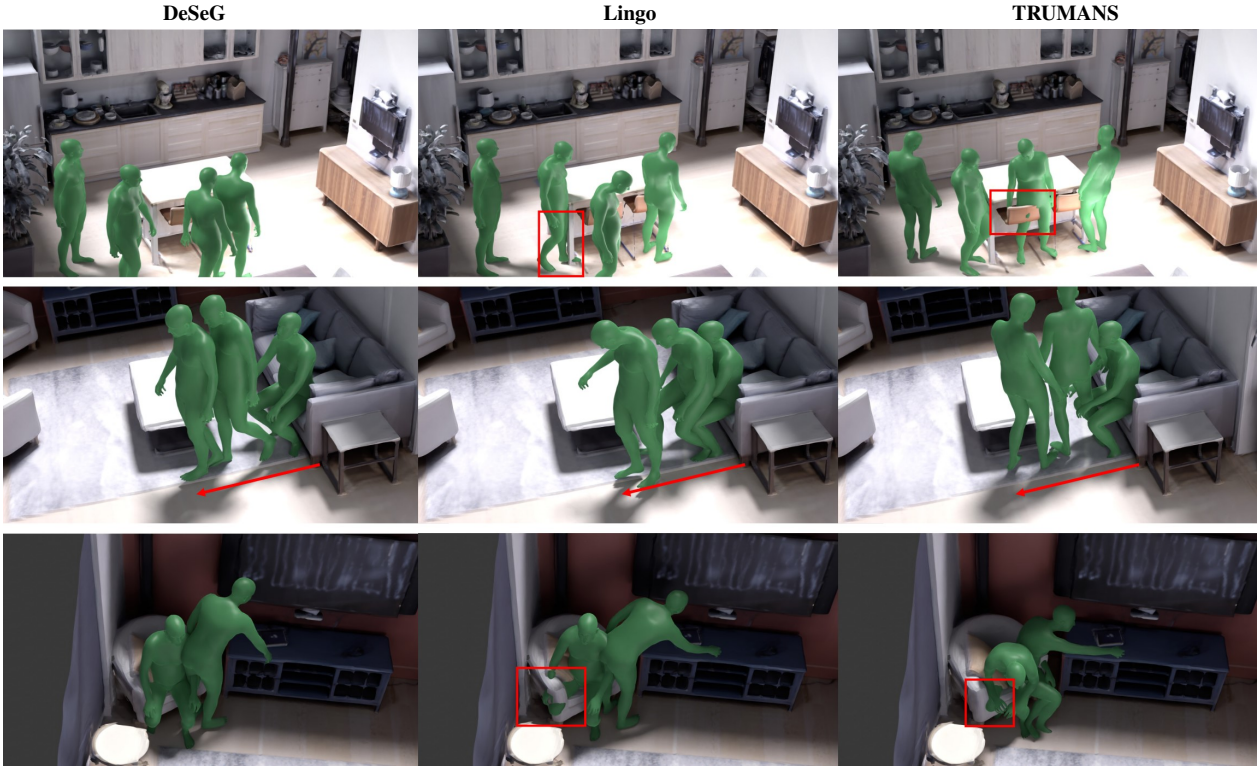


Figure 3. Qualitative comparison of long-horizon HSI sequences. **Rows** (top to bottom): *"walk around the table"*, *"stand up and creep"*, and *"pick up the book on cabinet"*. **Columns**: DeSeG, Lingo [15], and TRUMANS [16]. Red marks highlight contact and trajectories.

For $N = 30$ cases and $K = 3$ seeds, the automatic rates are

$$\begin{aligned} \widehat{p}_x &= \frac{1}{NK} \sum_{i=1}^{N} \sum_{s=1}^{K} S_{x,i,s}, \\ \widehat{p}_{pair} &= \frac{1}{NK} \sum_{i=1}^{N} \sum_{s=1}^{K} S_{can,i,s} S_{conf,i,s}. \end{aligned} \tag{13}$$

Pairing precedes averaging; $\widehat{p}_{pair}$ is not the product of the two aggregate success rates. Each method has 90 matched pairs (180 motions), whereas SGC aggregates 150 human judgments.

Table 2. **Affordance selection on NC-Bench (%).** SGC aggregates 150 human judgments per method; automatic metrics use 90 case–seed outputs per condition and method. Higher is better except for Prior Capture. †Different Planner checkpoint.

Method	SGC↑	Can. Succ.↑	Conf. Succ.↑	Paired Succ.↑	Prior Capture↓
Mono.+Rep.	38.0	82.2	45.6	36.7	40.0
Legacy no-reg.†	62.0	90.0	65.6	63.3	18.9
Lingo	58.0	84.4	61.1	58.9	21.1
DeSeG	**75.3**	**93.3**	**83.3**	**78.9**	**13.3**

4.2. Quantitative Results

Semantic Alignment and Motion Quality. As shown in Tab. 1, DeSeG achieves the best FID (2.882) and R-Precision (0.562) under the original LINGO protocol, while preserving high Diversity (5.865). Relative to the strongest prior result for each metric, DeSeG reduces FID by 6.8% (3.093 to 2.882) and improves R-Precision by 9.8% (0.512 to 0.562). Thus, its gains in instruction alignment are accompanied by competitive motion-distribution quality and diversity.

Physical Plausibility. DeSeG attains the lowest mean penetration, improving the strongest prior result from 0.278 to 0.207 (25.5%), while its maximum penetration of 0.921 is close to the best value of 0.915. Together with the FID and R-Precision gains, these results show improved scene compatibility without degrading semantic motion quality.

TRUMANS-derived Evaluation. On the TRUMANS-derived subset, DeSeG improves FID from 0.493 to 0.338, PFC from 0.657 to 0.621, and contact ratio from 0.941 to 0.986 relative to Lingo. Mean and maximum penetration decrease by 48.1% and 31.3%, respectively. These zero-shot results compare LINGO-trained models on the custom

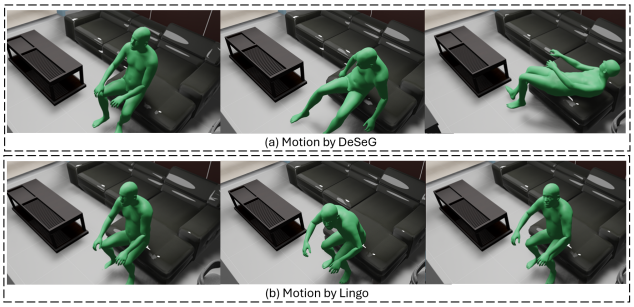


Figure 4. Qualitative result on a TRUMANS-derived scene for **sit down on sofa and then sleep**. DeSeG composes locomotion, sitting, and lying into a continuous scene-aware sequence.

subset, rather than the official TRUMANS evaluation split.

Results on NC-Bench. Tab. 2 further evaluates affordance selection. DeSeG improves canonical, conflict, and paired success over Monolithic+Repulsion from 82.2%/45.6%/36.7% to 93.3%/83.3%/78.9%. Prior Capture decreases from 40.0% to 13.3%, showing that the model is less likely to replace the instruction with an object's dominant interaction. Blinded SGC likewise rises from 38.0% to 75.3%. The simultaneous gains in canonical and conflict conditions show that hierarchical planning improves instruction-sensitive affordance selection while retaining object-compatible execution.

4.3. Model Configuration Comparisons

Hierarchical Planning. The available no-latent and no-cross variants obtain FID/R-Precision of 3.279/0.430 and 3.113/0.457, versus 2.882/0.562 for DeSeG. These configuration comparisons do not isolate the contribution of cross-attention, canonicalization, or the residual posterior. More directly, the matched-depth Monolithic+Repulsion model in Tab. 2 obtains only 36.7% paired success, compared with 78.9% for DeSeG. Together, the conventional and paired evaluations support the explicit affordance-planning interface.

Physical Regularization. The available no-regularization variant records mean/maximum penetration of 0.367/1.393 versus 0.207/0.921 for DeSeG, while R-Precision drops from 0.562 to 0.514. Because its Planner checkpoint also differs, the comparison measures a configuration-level difference rather than the independent effect of the physical loss. The supplementary material specifies both checkpoint pairs and the training objective.

4.4. Qualitative Analysis

Visualization. As shown in Fig. 3, DeSeG follows multi-stage instructions while maintaining coherent trajectories around scene objects. Compared with Lingo and TRUMANS, the generated bodies exhibit closer alignment with the specified path and fewer visible contact artifacts. Fig. 4 further shows transfer to a TRUMANS-derived scene, where the model connects locomotion, sitting, and lying through continuous transitions.

Long-Horizon Composition. DeSeG composes action-level segments autoregressively: the last two joint frames become the fixed prefix of the following segment. This mechanism preserves short-term pose continuity and enables the multi-phase interactions shown in the supplementary videos.

Failure Modes. Remaining failures arise in tightly constrained or long-horizon sequences. Coarse occupancy can miss thin surfaces and narrow clearances, while joint-space generation and body-model recovery may produce head, neck, or contact artifacts despite low joint penetration. Autoregressive composition can also accumulate pose errors. Thus, the reported geometric metrics measure scene compatibility, but do not guarantee collision-free meshes or uniformly natural motion.

5. Conclusion

We presented DeSeG, a two-stage framework that separates target-centered affordance planning from body-centered motion execution. A residual CVAE encodes language, task goals, and geometry surrounding the target into a compact affordance latent, which conditions a scene-aware diffusion Executor. An occupancy-derived repulsive objective penalizes proximity to occupied regions during training. DeSeG achieves the best FID, R-Precision, and mean penetration among the evaluated LINGO baselines, and improves canonical and conflict success on the paired NC-Bench evaluation.

Limitations and Future Work. DeSeG targets static indoor scenes. Coarse occupancy limits fine contact modeling, and autoregressive generation can accumulate errors. NC-Bench covers only five object categories. Dynamic objects, broader affordance evaluation, and globally consistent long-horizon motion remain future work.

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